Which Data Would Reduce This Uncertainty? Acquisition-Attributed Epistemic Widths for Frozen Classifiers

Anonymous Author
Anonymous Institution

Abstract

Epistemic uncertainty is uncertainty that more data can reduce, but which data? For a frozen classifier with a labeled reference set and additional unlabeled examples, we provide an estimated per-input answer. From the log we form a maximum mean discrepancy ball of input densities that the log cannot exclude, spanning prevalence mixtures of the known classes plus a slot for an unseen class. The spread of the induced label posteriors defines three epistemic widths: prevalence, label and coverage, each tied to a different acquisition. Under frozen class-conditionals, we characterize how class distinguishability and support determine when more logging narrows the prevalence and coverage widths, while the label width, fixed by the labeled reference, remains unchanged. In controlled synthetic experiments that vary log size, labeled sample size, and label noise, we test these responses, compare prevalence estimation with an EM-based bootstrap baseline, and check ball inclusion together with informativeness.

1 INTRODUCTION

Epistemic uncertainty is said to be what more data would reduce, but the phrase leaves open *which* data. Most deployments can buy three kinds: more unlabelled inputs from the deployment stream, more labelled examples of the classes already known, and labelled examples of a class not seen before. A single scalar does not say which of these would help at a given input. We make the acquisition explicit for a setting practitioners already have: a classifier trained once and then frozen, the labelled data it was trained against, and a growing unlabelled log of the inputs it meets in deployment. From the log we form the set of deployment distributions that are mixtures of the known classes (plus one candidate novel source) and lie within a computable distance of what the log shows, and the spread of the label posteriors this set admits splits into three parts: prevalence, label, and coverage of novelty.

The split matters because the parts answer to different acquisitions. When the class distributions are known, uncertainty about how common each class is falls only with more of the deployment stream, uncertainty about the label rates falls only with labels, and a novel source can be told apart from a change in known prevalences only when its inputs fall somewhere the known classes do not. When the class distributions are themselves estimated from labels, as in practice, labels and log both move the prevalence width, at an exchange rate the theory motivates: each acquisition acts through the part of the radius it controls. A score that merges everything, such as ensemble disagreement, moves with every acquisition that feeds it and cannot say which one helps (Section 5.2). Young [2026] argues the conceptual point, that reducibility is relative to an acquisition class; we supply estimators, finite-sample inclusion and tests of the acquisition decisions they imply, including one that fails.

Approach. The frozen classifier maps inputs to finitely many cells, so each class is a distribution over cells and the deployment law is a mixture of them. With the categorical kernel, the maximum mean discrepancy (MMD) between such laws is the Euclidean distance between cell histograms, so the set of mixtures consistent with the log is a ball whose radius has an explicit constant. Each width is the range of one posterior over that ball, computed without relaxation by a second-order cone program. When the class distributions are themselves estimated from labels, we widen the intervals to range over their estimation er-

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ror as well, and these joint intervals carry the coverage claim.

Contributions.

- Three widths with distinct estimands (prevalence, label, coverage) and joint intervals that keep finite-sample inclusion when class distributions are estimated, including a tight version that counts reference error once (Section 2).
- Conditions under which each width contracts, the exchange rate between log and labels they imply, and where coverage identification stops, with machine-checked cores (Section 3).
- A pre-registered synthetic study that varies log size, labeled sample size and label noise: with known rows each width responds to the acquisition it tracks, ensemble mutual information responds to all three, the ball includes the truth in every seed, and EMQ with a bootstrap loses inclusion where prevalence is not identified (Section 5.2).
- A CIFAR-10 audit with two frozen encoders against a finite-sample constraint region and a two-stage bootstrap (Section 5.3); a registered test showing that under shift the widths do not choose data better than a constant policy (Section 5.4); and a registered test on a blind encoder showing that, when the class-conditionals hold, the radius exchange rate picks the acquisition that tightens a certified interval more; on real features the gain over always buying more log is small, the choice is per deployment rather than per input, and data-free bookkeeping of the two radii does as well as the full method (Section 5.5).

2 SETUP

Frozen classifier and cells. A classifier trained once is never updated. It maps each input to one of d cells: in our experiments the predicted class of a linear-discriminant head on frozen features, plus one outlier cell for inputs far from every class mean. There are K known sources (classes, or groups of classes) with cell distributions $A_1, \dots, A_K \in \Delta_d$, and one slot for an unseen source with declared distribution $q \in \Delta_d$; stack them as the rows of $A \in \mathbb{R}^{(K+1)\times d}$. Deployment draws n i.i.d. cells from $p^* = w^{*\top} A$ with unknown $w^* \in \Delta_{K+1}$, and $\hat{p}_n$ is their histogram. Source k emits label $Y = 1$ with probability θ_k, independently of the cell given the source. The *frozen class-conditionals* assumption is that the rows A_k are the same in reference and deployment; label shift satisfies it, covariate shift in general does not. When the cells are the predicted classes, A is the head's confusion matrix and $w^\top A = \hat{p}_n$ is the moment equation of black-box shift estimation [Lipton et al., 2018]; the categorical kernel below adds no information beyond it. What we add is a set rather than a point, and its use.

The ball. With the categorical kernel $k(x, x') = \mathbf{1}\{x = x'\}$, the kernel mean embedding of a cell law is its probability vector and the maximum mean discrepancy (MMD) between two laws is their Euclidean distance; matching the log by a mixture of known embeddings is kernel mean matching [Zhang et al., 2013, Iyer et al., 2014]. We keep every mixture the log cannot exclude:

$$\mathcal{W}_n = \{\, w \in \Delta_{K+1} : \ \|w^\top \hat{A} - \hat{p}_n\|_2 \le \varepsilon_n + r \,\}, \quad (1)$$

with $\varepsilon_n = (1 + \sqrt{2\log(1/\alpha)})/\sqrt{n}$ (Lemma 1), $\hat{A}$ the rows estimated from m labelled reference images per source, and r their radius ($r = 0$, $\hat{A} = A$ when rows are known). The failure budget α is split between the log, the references and the label channel.

Three widths. For a probe cell x and source k, each width is the range of one posterior; they are distinct estimands, not an additive decomposition.

- *Prevalence* $W_{\mathrm{cls}}(x)$: range of $w_k A_k(x) / \sum_{j \le K} w_j A_j(x)$ over $w \in \mathcal{W}_n$.
- *Label* $W_{\mathrm{label}}(x)$: range of the label posterior at fixed anchor weights over the label-channel intervals, i.e. over whatever the labels estimate. In the synthetic study labels estimate the source label rates θ; on real data, where the label is the source, labels estimate the rows, and W_{label} ranges over the entries $A_k(x)$. The two instances share a role (the width that only labels narrow) but not an estimand, and Proposition 2 covers the first only.
- *Coverage* $W_{\mathrm{unk}}(x)$: range of $w_{K+1} q(x) / w^\top A(x)$ over $w \in \mathcal{W}_n$, the share of the probe that the known classes may fail to cover. The unseen source is a declared component, so this is closed-set inference with one extra row, not open-set detection.

Each endpoint is a linear-fractional program over the ball, a second-order cone program after the Charnes–Cooper change of variables [Charnes and Cooper, 1962]. A probe that some admissible mixture gives zero mass has an undefined posterior and receives $[0, 1]$ by policy; if the ball is empty the declared family is rejected and the method abstains.

Joint intervals. W_{cls} and W_{unk} evaluate the posterior at $\hat{A}$. A valid interval for the true posterior must also range over the row entries, which we bound by per-entry Hoeffding intervals $[\underline{A}_k(x), \overline{A}_k(x)]$ on their own share of α; for fixed weights the extremes sit at interval endpoints, so each side is again one cone program. The *joint* interval does this over the ball of radius $\varepsilon_n + r$, which counts reference error twice (in r and in the entries). The *tight* joint interval counts it once: it keeps the weights w for which the log histogram lies within ε_n of the box $\{x : \sum_j w_j \underline{A}_j(x) \le x_c \le \sum_j w_j \overline{A}_j(x)\}$, still a cone constraint and valid on the same event, since the true marginal lies in that box. This is the constraint region of Li et al. [2026] with its per-cell tolerance on the log replaced by the Euclidean MMD tolerance; it is "tight" only in counting reference error once, not in being sharp. Joint intervals carry *inclusion* of the target (we reserve "coverage" for the width W_{unk}); the three widths carry attribution.

From widths to an acquisition. At budget B, with one label costing c unlabelled inputs, REACH predicts how more log or more labels would change the joint width, holding current estimates fixed, and recommends the larger reduction; if the ball rejects the declared family or the probe is the outlier cell, it recommends labels from the deployment domain. The rule carries no guarantee: Section 5.4 tests it under shift and Section 5.5 when the class-conditionals hold.

3 WHEN EACH WIDTH CONTRACTS

Let $T = \{v \in \mathbb{R}^{K+1} : \mathbf{1}^\top v = 0\}$ be the tangent space of the simplex and

$$\sigma(A) = \min_{v \in T,\ \|v\|_2 = 1} \|v^\top A\|_2$$

the smallest singular value of the embedding map on weight changes: our measure of class distinguishability. $\sigma(A) > 0$ iff the rows of A are affinely independent, and $\sigma(A) \le s$, where $s = \operatorname{dist}(q, \operatorname{conv}\{A_k\}_{k \le K})$ separates the unseen slot from the known classes (Appendix A). So whenever the unseen row is in A, every bound below that needs $\sigma > 0$ also needs $s > 0$. Proofs are in Appendix A.

Lemma 1 (Ball inclusion)**.** *Under (L1) an i.i.d. log and (L2) the deployment law in the declared family, $p^* = w^{*\top} A$, McDiarmid's inequality gives $\Pr(\|\hat{p}_n - p^*\|_2 \le \varepsilon_n) \ge 1 - \alpha$. If also (L3) $\|\hat{A}_k - A_k\|_2 \le r$ for every row, then $w^* \in \mathcal{W}_n$, and every interval computed over $\mathcal{W}_n$, and over label-channel intervals that hold on their own events, contains its target.*

Proposition 1 (Prevalence contraction)**.** *On the event of Lemma 1, for every probe x,*

$$W_{\text{cls}}(x) \le L(x)\, \frac{2(\varepsilon_n + r)}{\sigma(\hat{A})}, \qquad L(x) = \frac{2\sqrt{K}\, \max_{k \le K} \hat{A}_k(x)}{\min_{w \in \mathcal{W}_n} \sum_{j \le K} w_j \hat{A}_j(x)},$$

provided (B′1) $\sigma(\hat{A}) > 0$ and (B′2) the probe keeps known mass over the ball. Since $\sigma(\hat{A}) \ge \sigma(A) - \sqrt{K+1}\, r$, the bound is $O(n^{-1/2})$ with known rows and $O(n^{-1/2} + m^{-1/2})$ with rows from m labels, as long as $\sigma(A)$ and the probe mass stay bounded away from zero.

Exchange rate. Holding $\sigma(\hat{A})$ and $L(x)$ fixed, the bound of Proposition 1 depends on the two acquisitions through the radius $\varepsilon_n + r_m$, with $\varepsilon_n \propto n^{-1/2}$ from the log and $r_m \propto m^{-1/2}$ from the references. At a budget B with label price c the bound's radius factor therefore falls more under the log iff $\varepsilon_n - \varepsilon_{n+B} > r_m - r_{m+B/(cK)}$, so which acquisition tightens the bound more changes with the starting budgets and the price. This is a property of the bound, not of the width, and it is data-free. Section 5.5 tests whether it nevertheless predicts which acquisition tightens a certified interval the proposition does not cover.

The constant shows where the proposition fails first: when rows are nearly affinely dependent ($\sigma \to 0$, which the resolution gate of Section 5.1 measures), and when a probe carries little known mass, which is why one source per CIFAR-10 class is close to vacuous for rare classes (Section 5.3). The $m^{-1/2}$ term is the cross-effect of labels on W_{cls} seen on real data.

Proposition 2 (Label-information limit)**.** *Suppose (B1) frozen class-conditionals with known rows and (B2) Y independent of the cell given the source. Then the law of the log does not depend on θ. For any $\theta \ne \theta'$ and any interval procedure that uses only the log and covers at level $1 - \alpha$ under both, both values lie in the interval with probability at least $1 - 2\alpha$, at every n; at a probe with known-source shares $\pi_k(x)$ the label-posterior interval therefore has width at least $|\sum_k \pi_k(x)(\theta_k - \theta'_k)|$ with that probability. With m labels per source and a known symmetric flip rate $\eta < 1/2$, the label-channel half-width is $O(\sqrt{\log(1/\alpha)/m}/(1 - 2\eta))$.*

Proposition 2 concerns θ, the synthetic instance of W_{label}. On real data W_{label} ranges over row entries, about which the log is not silent, so there its invariance to n holds by construction (fixed anchor weights) and is not evidence for Proposition 2. The proposition is the label-shift instance of the non-identification argument of Young [2026].

Proposition 3 (Coverage: where it contracts and where it stops)**.** *On the event of Lemma 1, for a probe x with $q(x) > 0$: (i) if (C1) $\sigma(\hat{A}) > 0$ with the unseen row included and (C2) the probe*

has positive mass under every $w \in \mathcal{W}_n$*, then* $W_{\mathrm{unk}}(x) \le L_{\mathrm{unk}}(x)\, 2(\varepsilon_n + r)/\sigma(\hat{A})$ *with* $L_{\mathrm{unk}}(x) = 2\sqrt{K+1}\max_j \hat{A}_j(x)/\min_{w\in\mathcal{W}_n} w^\top \hat{A}(x)$*, which vanishes as* $n, m \to \infty$*; (ii) if* $q \in \mathrm{conv}\{A_k\}$ *(*$s = 0$*) and* w^* *has slack, then at every* n *the ball contains a segment of weights with the same marginal and different unseen mass, so* $W_{\mathrm{unk}}(x)$ *is bounded below by a positive constant that does not shrink with* n*; (iii) if some* $w \in \mathcal{W}_n$ *gives the probe zero mass, the interval is* $[0, 1]$ *at every* n*.*

The same segment changes the mix of known sources, so at a probe with $q(x) > 0$ that two or more known sources share, the prevalence width (with true rows, or the joint interval) is generically bounded below as well; (B′1) fails because $\sigma \le s = 0$.

Support alone is therefore not enough: separation is needed too, measured in the encoder's cell space, and (iii) is a policy for an undefined posterior, not a theorem. The proposition takes q as declared; with only an admissible class $\mathcal{Q}$ of unseen distributions the interval must also range over $\mathcal{Q}$, and identification then needs conditions of the kind studied for mixture proportion estimation and open-set label shift [Ramaswamy et al., 2016, Garg et al., 2022]. The bounds above concern plug-in widths; we do not claim a rate for the joint intervals.

Machine-checked cores. We verified in Lean 4 with Mathlib the deterministic step of Lemma 1, the diameter bound $\sigma\|w - w'\| \le 2(\varepsilon_n + r)$ and the posterior Lipschitz bound behind Proposition 1, the two-point argument of Proposition 2, and the non-identification segment of Proposition 3(ii) (Appendix B); each depends only on Lean's standard axioms. Not formalised: the concentration inequality, the ℓ_1-to-ℓ_2 constant, and the modelling step from (B1)–(B2) to equal log laws, which the Lean statement takes as a hypothesis.

4 RELATED WORK

Prevalence under label shift. EMQ [Saerens et al., 2002], black-box shift estimation [Lipton et al., 2018], its regularised variant [Azizzadenesheli et al., 2019], calibrated maximum likelihood [Alexandari et al., 2020], the unified view of Garg et al. [2020], semiparametric estimation [Tian et al., 2023], MMD class-ratio estimation [Iyer et al., 2014] and distribution feature matching [Dussap et al., 2023] estimate prevalence under frozen class-conditionals; with hard predictions as cells our ball is a confidence set for the black-box moment system. Finite-sample sets for it exist: Si et al. [2024] propagate intervals through Gaussian elimination to build PAC prediction sets, Li et al. [2026] replace inversion by a constraint region solved by linear programs, account for an estimated confusion matrix, and bound the region's diameter, and Podkopaev and Ramdas [2021] give conformal guarantees under label shift. Our prevalence width is the same kind of object; Section 5.3 runs a constraint region after Li et al. [2026] side by side with ours. What we add is the label and coverage components, the attribution of each to an acquisition, and a test of that attribution.

Unseen classes. Under open-set label shift a new class arrives with unknown prevalence; Garg et al. [2022] identify its proportion under separability, building on mixture proportion estimation [Ramaswamy et al., 2016], and Liu et al. [2025] give asymptotic intervals. Our coverage width uses a *declared* unseen component and is closed-set: with q declared, separation $s > 0$ is all that Proposition 3 needs, whereas with q unknown identification requires conditions such as separability [Garg et al., 2022] or a density-ratio model [Liu et al., 2025].

Decomposing uncertainty and choosing data. The aleatoric/epistemic split [Hüllermeier and Waegeman, 2021] is usually measured by mutual information between parameters and prediction [Depeweg et al., 2018], estimated with ensembles [Lakshminarayanan et al., 2017]; Gruber et al. [2023] survey the sources of uncertainty from a statistician's view. Young [2026] argues that reducibility is relative to an acquisition class, separates uncertainty reducible by more of the same data from uncertainty reducible only by different data, and shows that ensemble disagreement need not track the epistemic term; our Proposition 2 is the label-shift instance of that argument. Our contribution relative to it is constructive: estimators with finite-sample inclusion for one deployment setting, their attribution to three acquisitions, and a test of whether that attribution chooses data well. Acquisition by information gain [Houlsby et al., 2011], including under label shift [Zhao et al., 2021], chooses which inputs to label for a fixed model class, and active testing [Kossen et al., 2021] chooses which points to label to evaluate a model; we choose between kinds of data (more log or more labels) at a price, and the quantity we value is the width of a certified interval rather than expected information. Detecting that the class-conditionals have moved is dataset-shift detection [Rabanser et al., 2019]; our incompatibility test is a conservative instance, and Section 5.4 shows that its power, not the widths, limits acquisition under shift.

Table 1: What each method provides. Guarantee: finite-sample inclusion at each budget. Acquisition: a rule for choosing between kinds of data (REACH: heuristic under shift, Section 5.4; radius-based under frozen class-conditionals, Section 5.5). †Prediction sets for labels, with exact guarantees under known importance weights. ‡Under frozen class-conditionals, a data-free comparison of radii; it applies to the constraint region as well.

	guarantee	prev.	labels	unseen	attribution	acquisition
EMQ + bootstrap	–	✓	–	–	–	–
BBSE / RLLS	–	✓	–	–	–	–
Si et al. [2024]	✓	✓	–	–	–	–
Podkopaev and Ramdas [2021]	sets†	–	–	–	–	–
Li et al. [2026]	✓	✓	✓	–	–	–
OSLS [Garg et al., 2022]	–	✓	–	✓	–	–
Ensemble MI	–	one scalar			–	–
REACH	✓	✓	✓	declared	✓	radius rule‡

5 EXPERIMENTS

We test four things: that each width responds to the acquisition it is built to track (Section 5.2), that the joint intervals include their targets on frozen real features (Section 5.3), whether the widths choose an acquisition that pays under shift (Section 5.4), and whether the exchange rate of Section 3 chooses the better acquisition when the class-conditionals hold (Section 5.5). Each test's expected outcomes were registered in the code before it ran; deviations are listed in Appendix E.

5.1 Protocol

Response slopes. For each acquisition we sweep its budget on a log-spaced grid with the other two at base values, and fit the least-squares slope of log(mean width) on log(budget), so $-1/2$ is the parametric rate. Confidence intervals bootstrap over seeds, the independent replicate. A slope is called "≈ 0" when it lies inside the $1-\alpha/K$ quantile ($\alpha = 0.05$, Bonferroni over the K registered cells) of slopes refitted on fresh same-budget seeds randomly assigned to grid levels. This is a failure to reject against a placebo null for that estimator at that seed count, not an equivalence test: it says the response is not distinguishable from Monte Carlo noise; the tolerance τ of every cell is in the released response matrices so its power is visible. Results are conditional on one fitted head and one test pool per encoder, and we make no multiplicity correction across sections. Invariances that hold by construction are marked 0^*. Before any slope is reported the pipeline checks that the ball resolves a 0.4 shift of weight between the two closest sources, $\|0.4(\hat{A}_i - \hat{A}_j)\|/(\varepsilon_n + r) > 1$, at the smallest n.

Baselines. EMQ [Saerens et al., 2002] with a multinomial bootstrap of the log gives percentile intervals for the same estimands as W_{cls} and W_{unk}, without a finite-sample guarantee. A bootstrap ensemble that resamples both the log and the labels gives ensemble mutual information $\mathrm{H}(\bar{p}) - \overline{\mathrm{H}(p)}$, the scalar score the introduction argues against.

Table 2: Response matrix, world *separated*, on-support probe: log-log slope of mean width against each acquisition. Arrows: outside the placebo band; 0^*: invariant by construction; the W_{label} arrows follow from its closed-form radius. Bootstrap half-widths < 0.03.

	unlabelled log n	labels m	noise η
W_{cls}	$-0.51\downarrow$	0^*	0^*
W_{label}	0^*	$-0.47\downarrow$	$+0.16\uparrow$
W_{unk}	$-0.49\downarrow$	0^*	0^*
EMQ+bootstrap	$-0.50\downarrow$	0^*	0^*
Ensemble MI	$-0.03\downarrow$	$-0.82\downarrow$	$+0.44\uparrow$

5.2 Synthetic response matrix

Three sources over four cells: two known sources and a novel one with mass $\delta = 0.1$, label rates $\theta = (0.2, 0.8, 0.9)$. Probe cells are *on support*, *boundary* (known-source overlap) and *off support* (zero mass). The novel source lies outside the known hull in world *separated*, at the midpoint of the known rows in *confounded*, and is absent in *absent*. Grids are $n \in \{2^{10}, \ldots, 2^{16}\}$, $m \in \{2^5, \ldots, 2^{11}\}$, $\eta \in \{0.0125, \ldots, 0.2\}$, 128 seeds per budget; the resolution gate reads 2.44.

Table 2 shows the predicted pattern. W_{cls} contracts with the log at the parametric rate (slope -0.51, 95% CI $[-0.508, -0.503]$), W_{label} contracts with labels (-0.47) and widens with noise ($+0.16$), and W_{unk} contracts with the log (-0.49) on support and stays at 1 off support. Ensemble MI responds to all three (-0.03, -0.82, $+0.44$), dominated by the label channel: on these probes it barely registers the log, although the prevalence width falls eightfold over the same grid. Of the 25 registered cells, 12 concern the baselines (all agree) and 13 concern REACH. Of those, two are off-support cells that are $[0, 1]$ by policy and four follow from the closed-form label radius, so neither group is evidence; of the remaining seven, five agree (prevalence and coverage contraction) and two do not. Both failures are the coverage width in *confounded*, registered as no response: the width stays large ($1.0 \to 0.69$ on support, 0.86 on the boundary at $n = 2^{16}$) but falls with slope -0.09 and -0.04, as the ball shrinks onto a wide identified set. "Attenuated fivefold, bounded below", as Proposition 3(ii) states, would have been the right registration. Prevalence also contracts more slowly there (slope -0.35 on support, and -0.08 on the boundary, $0.93 \to 0.63$): with the

Table 3: CIFAR-10 label shift, CLIP, $n = 16{,}000$, $m = 2000$, 50 seeds: per-probe inclusion of the true posterior (column "coverage") and mean width on known-class cells.

	4 superclass sources		10 class sources	
	coverage	width	coverage	width
REACH joint interval	1.00	0.71	1.00	0.88
Constraint region (after Li et al.)	1.00	0.65	1.00	0.74
EMQ, two-stage bootstrap	0.82	0.06	0.58	0.07
EMQ, log-only bootstrap	0.13	0.01	0.17	0.03
REACH W_{cls} (plug-in)	0.98	0.46	1.00	0.76

unseen row inside the hull $\sigma = 0$, so Proposition 1 gives no contraction, and the remark after Proposition 3 bounds the shared-probe width below. The ball contains the true marginal in every seed at every n (Clopper–Pearson lower bound 0.977 at nominal 0.975). EMQ+bootstrap is about three times narrower and near nominal in *separated* (0.92–0.96), but in *confounded* its coverage falls from 0.73 to 0: EM picks one point of a non-identified set and the bootstrap shrinks around it (Figure 1, left).

5.3 CIFAR-10 audit with frozen encoders

CIFAR-10 images [Krizhevsky, 2009] pass through a frozen CLIP ViT-B/32 [Radford et al., 2021]; PCA to 32 dimensions and an equal-prior Gaussian LDA head are fitted on 10,000 training images (accuracy 91.2%). Rows are estimated from m of the remaining 40,000 per source, and the log is drawn from the test set. With one source per class the ball cannot resolve classes whose deployment mass is below its radius (about 0.10 at $m = 2000$), so the headline uses four superclass sources with fixed within-group composition (which makes frozen class-conditionals hold by construction for label shift); ten sources are reported alongside. Both pass the resolution gate (1.14, 1.62). Label-shift and held-out arms use 50 seeds and Dirichlet(1) prevalences drawn per seed; coverage is judged at the solver's accuracy (10^{-6}). Two interval baselines join EMQ: a constraint region after Li et al. [2026], built from per-cell intervals on the log frequencies and the same per-entry intervals on the rows, with no ball; and a two-stage bootstrap that resamples the references as well as the log.

Coverage and width. Both finite-sample constructions include the target in every seed at every n, for both groupings (Table 3; the label-shift ball has the K known rows and no unseen slot), and at $n = 16{,}000$ the constraint region is narrower than or equal to REACH's joint interval on every probe (strictly on 66% and 58%; at $n \leq 4000$ it is wider on 6–16% of super4 probes), on average 0.65 against 0.71 with super4 sources and 0.74 against 0.88 with ten, although it spends a smaller share of the failure budget. Neither is informative in absolute terms: with super4 sources 41% of joint intervals are $[0, 1]$ and 30% exclude 0.5; with ten sources 68% are $[0, 1]$ and 13% exclude 0.5. On the grid of Section 5.5, the tight joint interval of Section 2 closes 56–71% of the gap between the joint interval and the region but remains wider than the region on most episodes. The bootstrap baselines are narrow and do not cover. Log-only EMQ covers 0.54 at $n = 500$ and 0.13 at $n = 16{,}000$, because reference error is ignored and the interval shrinks onto a biased point (Figure 1, centre); resampling the references as well lifts it to 0.82 (0.58 with ten sources). Plug-in W_{cls} includes the target in 0.98 of probes here and 0.94 with DINOv2 features, and carries no guarantee, which is why the joint interval carries the claim.

Attribution. All 13 registered cells agree for super4 (12 of 13 with ten sources, where the joint width is already saturated under noise). W_{label} does not read the log by construction, contracts with labels (-0.31) and widens with noise ($+0.21$). The joint width responds to all three acquisitions, -0.07 in n, -0.11 in m and $+0.04$ in η, so on real data the one-to-one attribution of the synthetic study does not hold for the interval that carries coverage: labels also narrow W_{cls} (-0.19) and label noise widens it ($+0.13$), because labels estimate the rows the ball is built from. What the theory predicts on real data is not one-to-one attribution but the exchange rate between the two radii, which Section 5.5 tests. Log-only EMQ's response to labels lies inside the placebo band although its interval excludes zero ($+0.06$, $[0.01, 0.11]$); ensemble MI responds to the log and to labels but not to label noise.

Unseen classes and corruption. Removing ship and truck from the labelled set ($s = 0.22$) and adding them as $\delta = 0.1$ of the log, or not at all, the joint interval for the coverage width includes its target in every seed and contracts in both arms ($0.92 \to 0.72$ present, $0.89 \to 0.66$ absent); plug-in W_{unk} under-covers with DINOv2 features (0.88), as Section 2 anticipates. A declared uniform unseen component is never flagged by the residual test, so we make no claim that the audit detects a wrong unseen model. On CIFAR-10-C [Hendrycks and Dietterich, 2019] (20 seeds per corruption and severity) the rows move. REACH abstains when the ball is empty, answering 94% of super4 runs at severity 1 and 45% at severity 5 and including the target in 0.998 and 0.93 of those it answers (0.94 and 0.42 unconditionally); the constraint region answers fewer (26%) and includes less (0.91), and bootstrap EMQ includes 0.21. Ten super4 and 21 ten-source corruption-severity cells under-cover, mostly with the test silent (Appendix D). A self-supervised DINOv2

Table 4: Mean regret (nats per cell) by policy. Left: price 1:50 by shift family; right: all 77 splits at prices 1:50 and 1:250. Evaluation seeds; prior strength $M_0 = 50$.

	price 1:50, by family			all splits	
policy	clean	C10.1	C10-C	1:50	1:250
REACH-v2	0.0029	0.0101	0.0864	0.0843	0.0411
flag only	0.0026	0.0096	0.0877	0.0856	0.0429
always labels	0.0086	0.0188	0.0100	0.0101	0.0084
always reweight	0.0019	0.0074	0.2382	0.2321	0.1189
do nothing	0.0018	0.0083	0.2401	0.2340	0.1208

encoder [Oquab et al., 2024] reproduces every pattern, and a CIFAR-100 arm locates where Proposition 3 stops (Appendix D).

5.4 Does the width comparison choose the data that pays?

Without ground-truth widths this test is behavioural: take the acquisition REACH recommends per cell and measure what it buys on held-out shifted data. Shift splits are clean CIFAR-10, CIFAR-10.1 [Recht et al., 2018] and the 75 CIFAR-10-C splits, each cut into disjoint log, label and evaluation pools. An episode draws Dirichlet(1) prevalences over the superclasses and starts from clean rows ($m = 2000$) and 1000 shifted log images. With budget 5000 units, *reweight* grows the log to 6000 and refits EMQ; *add labels* buys $5000/c$ shifted labels and shrinks the rows towards them ($M_0 = 50$); *do nothing* has payoff 0. Payoff is the reduction of held-out negative log-likelihood (nats) in a cell; regret is the best action's payoff minus the chosen one's. We compare REACH (Section 2) with every constant policy, the incompatibility flag alone, and ensemble MI mapped to actions by tercile with the map tuned on calibration-seed regret. Seeds 10–19 are evaluated; intervals use a two-way cluster bootstrap over seeds and shift families. This is the second registered version of the test: a first version's rule had two defects that review exposed (a non-finite predicted gain that defaulted to labels, and a label gain priced for clean rather than shifted labels), and its comparison tuned the baseline for accuracy rather than regret (Appendix E).

Result. The registered prediction fails: buying labels every time has lower regret than REACH at every price (0.010 against 0.084 at 1:50, difference CI $[0.038, 0.134]$; 0.008 against 0.041 at 1:250, $[0.014, 0.063]$) and at every prior strength $M_0 \in \{10, 50, 200\}$, and the regret-tuned MI map is the same constant policy. The reason follows from the registered rule itself: under frozen class-conditionals the predicted label gain can only widen the rows' radius when $n_k < m$ (true at every price), so in a compatible frame REACH never predicts that labels help (largest predicted label gain 3×10^{-8} at 1:50, 6×10^{-8} over all settings). Its "label-limited" cells are ties on vacuous intervals ($W = 1$), and 15% of the mass falls to "do nothing", largely by ties as well; the oracle has zero regret by definition. All of its label recommendations therefore come from the incompatibility test, which fires in only 33% of severity-5 episodes, while shifted labels pay 0.23 nats under corruption. On clean CIFAR-10 and CIFAR-10.1, where labels slightly hurt (-0.007 and -0.011 nats), REACH beats always-labels but not doing nothing, always reweighting or the flag alone (Table 4). Beyond the flag the width comparison adds nothing: REACH minus flag-only lies within $[-0.005, 0.0005]$ at 1:50, and the registered contrast H1u (labels' advantage where REACH recommends them, unflagged cells) is -0.049 ($[-0.101, 0.019]$) at 1:50, negative in three of five settings, and significantly negative at $M_0 = 10$ (-0.078, $[-0.152, -0.020]$). Fixing the version-1 defects raised REACH's regret at 1:50 from 0.032 to 0.084 (version 1 bought labels by accident). Under shift the widths cannot tell whether the rows have moved.

5.5 When the class-conditionals hold: the exchange rate chooses the data

Under frozen class-conditionals the question is which acquisition tightens a certified interval more at a given price. We registered the test before writing its code and before a third encoder existed: $n_0 \in \{500, 2000, 8000\}$, $m_0 \in \{125, 500, 2000\}$ ($\{32, 128, 512\}$ in a synthetic world with estimated rows), prices $c \in \{10, 50, 250\}$, budget 5000; the outcome is the realised reduction of the mean tight joint width after actually acquiring. REACH predicts each reduction by re-solving the tight interval at the post-acquisition radii; the data-free alternative compares the radius reductions alone. Primary settings are the synthetic world and a frozen ResNet-50 encoder [He et al., 2016] never inspected before the run; CLIP replicates on seen features; 30 seeds per base point, seeds 10–29 evaluated.

Result. All registered claims hold (Table 5), but what they establish is narrower than the registration suggests. The better acquisition changes across base points and prices in 14 of 27 synthetic cells and 4 of 27 real ones (labels win when the log is already large, references are few and labels cheap), and REACH picks it, with regret below both constant policies and the tuned MI map (CIs exclude zero in all three settings). Four qualifications. First, the decision value is the radius exchange rate's: the data-free comparison of radii

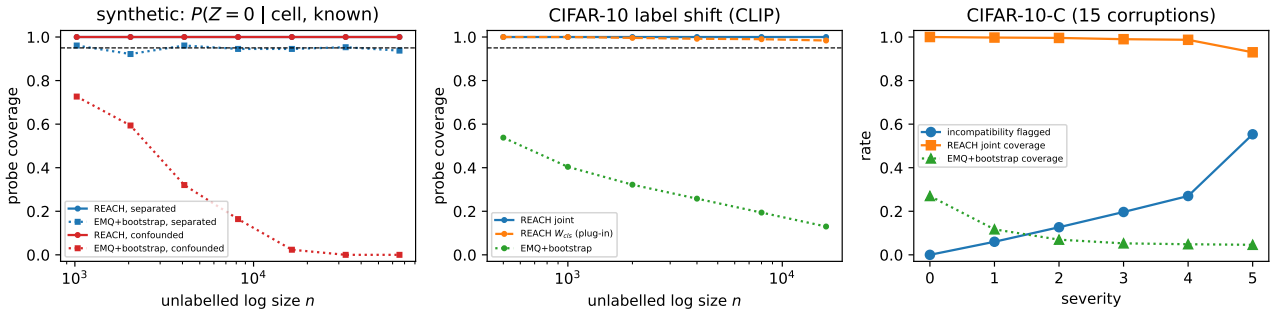


Figure 1: Per-probe inclusion of REACH and EMQ+bootstrap. Left: synthetic; both REACH curves lie on 1.0. Centre: CIFAR-10 label shift, four superclass sources. Right: CIFAR-10-C pooled over 15 corruptions, inclusion among runs where the ball is not empty, with the rate at which the incompatibility test fires. Constraint-region and two-stage-bootstrap results are in the text. Dashed: 0.95.

Table 5: Exchange-rate test under pure label shift: which acquisition tightens the certified interval more. Regret in width units, evaluation seeds; CI: seed bootstrap.

	synthetic (blind)	ResNet-50 (blind)	CLIP (seen)
cells where labels are better	14/27	4/27	4/27
predicted sign correct	1.00	1.00	1.00
corr. predicted vs realised gain	0.995	0.999	1.000
regret: REACH	0.0019	0.0001	0.0001
radius bookkeeping	0.0019	0.0000	0.0001
always log	0.0540	0.0032	0.0035
always labels	0.0487	0.0411	0.0426
ensemble MI (tuned)	0.0224	0.0032	0.0035
REACH − best constant, 95% CI	$[-0.0514, -0.0424]$	$[-0.0036, -0.0026]$	$[-0.0040, -0.0028]$
tight width (vs. region)	0.83 (0.79)	0.90 (0.89)	0.89 (0.88)
tight ≤ joint, share of probes	1.00	1.00	1.00
tight inclusion (min over bases)	1.00	1.00	1.00

makes the same choice in 93–94% of episodes, with regret equal to or lower than REACH's, and the same rule predicts the better acquisition for the constraint region in 25–26 of 27 cells; REACH's per-input widths add nothing to the choice, which is made per deployment, not per input. Second, on real features the stakes are small: "always more log" is already right in 23 of 27 cells (so it would pass E1 too), trails by 0.003 width units on intervals about 0.9 wide, labels win by 0.007–0.05, and three of the four cells where they win start near-vacuous; 29–32% of exchange episodes have mean width ≥ 0.999, and on real features the tuned MI map is "always log". Third, the correlation of 0.995–1.000 is over cell means (both actions pooled) and driven by the design; within cells it is 0.92–0.97 on real features and 0.37 for the synthetic label action, and the prediction re-solves the same program, so it is partly a self-consistency check. Fourth, the tested width is the tight joint interval, which Proposition 1 does not cover; for plug-in W_{cls}, the width the proposition bounds, the radius rule's sign is right in 20 of 27 synthetic cells and 25–26 of 27 real ones. The tight interval is never wider than the joint interval (strictly narrower in 69–72% of episodes) and includes the target in every seed, but remains wider than the constraint region (0.90 against 0.89 on ResNet-50, 0.83 against 0.79 in the synthetic world). In short: when the class-conditionals hold, bookkeeping of the two radii, which the theory motivates, chooses the acquisition well, and the per-input widths do not improve on it.

6 LIMITATIONS AND OUTLOOK

REACH estimates no aleatoric uncertainty, and its guarantees concern finitely many cells of a frozen classifier, hold per budget, and rest on the declared mixture family. On real features its intervals are wide, often $[0, 1]$, and a per-cell constraint region is at least as narrow; our tight interval is that region with a Euclidean log tolerance. What REACH adds is the acquisition-indexed decomposition, its conditions, and synthetic evidence for its attribution, not tighter intervals or a better acquisition rule. With estimated class distributions attribution becomes an exchange rate between radii, and when the class-conditionals hold, data-free bookkeeping of that rate chooses the acquisition as well as the full method, with small stakes on real features. Under shift the rule fails against a constant policy: its label arm cannot foresee the bias shifted labels correct, and the incompatibility test is too weak. The unseen component is declared, several design choices followed exploratory failures (Appendix E), and our real data are one image family with three encoders. A more powerful test that class-conditionals have moved, and pricing labels or routing non-contracting inputs to a human, are the open problems.

AI use statement

In this work, we used generative AI tools to write and run experiment and analysis code, to help design and write the pre-registration files, to draft Lean formalisations of the deterministic proof steps, to search for and check related work, to run simulated peer reviews of drafts, and to draft and edit the text. We did not use generative AI tools to generate the research question or the frozen abstract. We have reviewed all AI-assisted work: every reported number is regenerated from the released scripts and result files, the Lean statements are machine-checked with only Lean's standard axioms, every citation was checked against the cited work, and the authors checked the proofs in the appendix. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

REFERENCES

Amr Alexandari, Anshul Kundaje, and Avanti Shrikumar. Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation. In *International Conference on Machine Learning*, pages 222–232, 2020.

Kamyar Azizzadenesheli, Anqi Liu, Fanny Yang, and Animashree Anandkumar. Regularized learning for domain adaptation under label shifts. In *International Conference on Learning Representations*, 2019.

Abraham Charnes and William W. Cooper. Programming with linear fractional functionals. *Naval Research Logistics Quarterly*, 9(3–4):181–186, 1962.

Stefan Depeweg, Jose-Miguel Hernandez-Lobato, Finale Doshi-Velez, and Steffen Udluft. Decomposition of uncertainty in Bayesian deep learning for efficient and risk-sensitive learning. In *International Conference on Machine Learning*, pages 1184–1193, 2018.

Bastien Dussap, Gilles Blanchard, and Badr-Eddine Chérief-Abdellatif. Label shift quantification with robustness guarantees via distribution feature matching. *arXiv preprint arXiv:2306.04376*, 2023.

Saurabh Garg, Yifan Wu, Sivaraman Balakrishnan, and Zachary Lipton. A unified view of label shift estimation. In *Advances in Neural Information Processing Systems*, volume 33, pages 3290–3300, 2020.

Saurabh Garg, Sivaraman Balakrishnan, and Zachary C. Lipton. Domain adaptation under open set label shift. In *Advances in Neural Information Processing Systems*, 2022.

Cornelia Gruber, Patrick Oliver Schenk, Malte Schierholz, Frauke Kreuter, and Göran Kauermann. Sources of uncertainty in supervised machine learning – a statisticians' view. *arXiv preprint arXiv:2305.16703*, 2023.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2016.

Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations*, 2019.

Neil Houlsby, Ferenc Huszár, Zoubin Ghahramani, and Máté Lengyel. Bayesian active learning for classification and preference learning. *arXiv preprint arXiv:1112.5745*, 2011.

Eyke Hüllermeier and Willem Waegeman. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110(3):457–506, 2021.

Arun Iyer, Saketha Nath, and Sunita Sarawagi. Maximum mean discrepancy for class ratio estimation: Convergence bounds and kernel selection. In *International Conference on Machine Learning*, pages 530–538, 2014.

Jannik Kossen, Sebastian Farquhar, Yarin Gal, and Tom Rainforth. Active testing: Sample-efficient model evaluation. In *International Conference on Machine Learning*, 2021.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017.

Mushan Li, Kihyun Han, and Yanyuan Ma. From matrix inversion to constraints: Provably tighter confidence regions for importance weights in label shift. *arXiv preprint arXiv:2609.14802*, 2026.

Zachary Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and correcting for label shift with black box predictors. In *International Conference on Machine Learning*, pages 3122–3130, 2018.

Siyan Liu, Yukun Liu, Qinglong Tian, Pengfei Li, and Jing Qin. Semiparametric learning from open-set label shift data. *arXiv preprint arXiv:2509.14522*, 2025.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa,

Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

Aleksandr Podkopaev and Aaditya Ramdas. Distribution-free uncertainty quantification for classification under label shift. In *Conference on Uncertainty in Artificial Intelligence*, 2021.

Stephan Rabanser, Stephan Günnemann, and Zachary C. Lipton. Failing loudly: An empirical study of methods for detecting dataset shift. In *Advances in Neural Information Processing Systems*, 2019.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021.

Harish G. Ramaswamy, Clayton Scott, and Ambuj Tewari. Mixture proportion estimation via kernel embedding of distributions. In *International Conference on Machine Learning*, 2016.

Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do CIFAR-10 classifiers generalize to CIFAR-10? *arXiv preprint arXiv:1806.00451*, 2018.

Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41, 2002.

Wenwen Si, Sangdon Park, Insup Lee, Edgar Dobriban, and Osbert Bastani. PAC prediction sets under label shift. In *International Conference on Learning Representations*, 2024.

Qinglong Tian, Xin Zhang, and Jiwei Zhao. ELSA: Efficient label shift adaptation through the lens of semiparametric models. In *International Conference on Machine Learning*, pages 34120–34142, 2023.

Robin Young. Epistemic uncertainty is not the reducible kind. *arXiv preprint arXiv:2606.12646*, 2026.

Kun Zhang, Bernhard Schölkopf, Krikamol Muandet, and Zhikun Wang. Domain adaptation under target and conditional shift. In *International Conference on Machine Learning*, 2013.

Eric Zhao, Anqi Liu, Animashree Anandkumar, and Yisong Yue. Active learning under label shift. In *International Conference on Artificial Intelligence and Statistics*, 2021.

CHECKLIST

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. Yes (Sections 2 and 3).
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. Yes: each interval endpoint is one cone or linear program in $K+1$ variables with d cells, and the radii give the sample-size dependence (Section 2 and Lemma 1, Appendix C).
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. Yes (supplementary material).

2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. Yes (Section 3).
   (b) Complete proofs of all theoretical results. Yes (Appendix A); deterministic and measure-theoretic cores are also machine-checked in Lean (Appendix B).
   (c) Clear explanations of any assumptions. Yes (Sections 2 and 3).

3. For all figures and tables that present empirical results, check if you include:
   (a) The code, data, and instructions needed to reproduce the main experimental results. Yes (supplementary code, pre-registration files and commands; public datasets).
   (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). Yes (Section 5, Appendix C); nothing is trained beyond a PCA and LDA head on a fixed development split, and every design choice made after seeing results is listed in Appendix E.
   (c) A clear definition of the specific measure or statistics and error bars. Yes: bootstrap intervals over seeds, or over seeds and shift families (Section 5).
   (d) A description of the computing infrastructure used. Yes (Appendix C).

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
   (a) Citations of the creator if your work uses existing assets. Yes (CIFAR-10, CIFAR-10-C, CIFAR-10.1, CIFAR-100, CLIP, DINOv2, ResNet-50).

(b) The license information of the assets, if applicable. Yes, in the supplementary README.

(c) New assets either in the supplemental material or as a URL, if applicable. Yes (code, pre-registration files, Lean project).

(d) Information about consent from data providers/curators. Not Applicable.

(e) Discussion of sensible content if applicable. Not Applicable.

5. If you used crowdsourcing or conducted research with human subjects, check if you include:

(a) The full text of instructions given to participants and screenshots. Not Applicable.

(b) Descriptions of potential participant risks, with links to IRB approvals if applicable. Not Applicable.

(c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. Not Applicable.

A PROOFS

Distinguishability facts. (a) $\sigma(A) > 0$ iff no nonzero $v \in T$ has $v^\top A = 0$, i.e. iff the rows are affinely independent. (b) $\sigma(A) \le s$: let $c^* \in \Delta_K$ attain $s = \|q - \sum_k c_k^* A_k\|$ and $v = (e_{K+1} - c^*)/\|e_{K+1} - c^*\|$, a unit tangent vector; then $\|v^\top A\| = s/\sqrt{1 + \|c^*\|^2} \le s$. (c) For any unit $v \in T$, $\|v^\top \hat{A}\| \ge \|v^\top A\| - \sum_k |v_k| \|\hat{A}_k - A_k\| \ge \sigma(A) - \sqrt{K+1}\, r$.

Lemma 1. Write $\hat{p}_n = \frac{1}{n}\sum_i e_{X_i}$. By Jensen, $\mathbb{E}\|\hat{p}_n - p^*\|_2 \le (\mathbb{E}\|\hat{p}_n - p^*\|_2^2)^{1/2} = ((1 - \|p^*\|_2^2)/n)^{1/2} \le n^{-1/2}$. Replacing one draw moves $\hat{p}_n$ by at most $\sqrt{2}/n$ in ℓ_2, so by McDiarmid's inequality $\Pr(\|\hat{p}_n - p^*\|_2 \ge n^{-1/2} + t) \le \exp(-nt^2)$; the choice $t = \sqrt{\log(1/\alpha)/n}$ suffices, and the stated ε_n with $\sqrt{2\log(1/\alpha)}$ is larger by a factor $\sqrt{2}$ on the deviation term, hence also valid. On this event and (L3), for $w^* \in \Delta_{K+1}$, $\|w^{*\top}\hat{A} - \hat{p}_n\| \le \|\sum_k w_k^*(\hat{A}_k - A_k)\| + \|p^* - \hat{p}_n\| \le \sum_k w_k^* r + \varepsilon_n = \varepsilon_n + r$ (Lean: `ball_contains_truth`). Every interval over $\mathcal{W}_n$ then contains the value at w^*; intervals that also range over label-channel or component intervals contain the target on the intersection of their events, whose failure probability is bounded by the union bound used to split α.

Proposition 1. For $w, w' \in \mathcal{W}_n$, $v = w - w' \in T$ and $\|v^\top \hat{A}\| \le \|w^\top \hat{A} - \hat{p}_n\| + \|w'^\top \hat{A} - \hat{p}_n\| \le 2(\varepsilon_n + r)$, so $\sigma\|v\|_2 \le 2(\varepsilon_n + r)$ (Lean: `diam_bound`). At a probe, $f(w) = u(w)/v(w)$ with $u = w_k a_k$, $v = \sum_{j \le K} w_j a_j$, $a_j = \hat{A}_j(x) \in [0, a_{\max}]$. For $0 \le u' \le v'$ and $v, v' > 0$, $|u/v - u'/v'| \le (|u - u'| + |v - v'|)/v$ (Lean: `ratio_lipschitz`), and $|u - u'|, |v - v'| \le a_{\max}\|w - w'\|_1$, so $|f(w) - f(w')| \le 2a_{\max}\|w - w'\|_1 / v(w)$ (Lean: `posterior_lipschitz`). With $\|\cdot\|_1 \le \sqrt{K}\|\cdot\|_2$ on the K known coordinates, the only ones that enter f, and $v(w) \ge \min_{\mathcal{W}_n} v$, the width is at most $L(x)\, 2(\varepsilon_n + r)/\sigma$. The rates follow from $\varepsilon_n = O(n^{-1/2})$ and $r = O(m^{-1/2})$.

Proposition 2. Under (B1)–(B2) the log's law is that of n i.i.d. draws from $w^{*\top}A$, in which θ does not appear, so it is the same law P under θ and θ'. Let $C(\cdot)$ be a procedure that uses only the log with $P(\vartheta \notin C) \le \alpha$ for $\vartheta \in \{\theta, \theta'\}$. Then $P(\theta \in C, \theta' \in C) \ge 1 - P(\theta \notin C) - P(\theta' \notin C) \ge 1 - 2\alpha$, by subadditivity of outer measure, so no measurability of C is needed. On that event the interval has width at least $|\theta_k - \theta'_k|$, and for the label posterior at fixed shares $\pi_k(x)$ at least $|\sum_k \pi_k(x)(\theta_k - \theta'_k)|$ (Lean: `two_point`, `width_ge`; the Lean statement assumes coverage at every parameter value, which is stronger than at two). For the rate, each source's noisy label mean has Hoeffding half-width $\sqrt{\log(2K/\alpha)/(2m)}$ and debiasing a known symmetric flip divides it by $1 - 2\eta$.

Proposition 3. (i) As for Proposition 1 with $u = w_{K+1} q(x)$ and $v = w^\top \hat{A}(x)$, using (C1) for the diameter and (C2) for the denominator. (ii) Let $q = \sum_k c_k A_k$ with $c \in \Delta_K$, extended by $c_{K+1} = 0$. For $w_t = w^* + t(e_{K+1} - c)$, $w_t^\top A = w^{*\top} A$ for all t and $(w_t)_{K+1} = w^*_{K+1} + t$ (Lean: `novel_not_identified`); $w_t \in \Delta_{K+1}$ for $0 \le t \le \min_{k: c_k > 0} w_k^*/c_k$, which is positive when w^* has slack (Lean: `novel_segment_in_simplex`). On the event of Lemma 1, $\|w_t^\top \hat{A} - \hat{p}_n\| \le \|w_t^\top(\hat{A} - A)\| + \|p^* - \hat{p}_n\| \le r + \varepsilon_n$, so every w_t lies in $\mathcal{W}_n$. For $q(x) > 0$ the coverage posterior $w_{K+1} q(x)/w^\top A(x)$ is strictly increasing along the segment (the denominator is constant), so $W_{\mathrm{unk}}(x)$ is at least its increase over the segment, a positive constant independent of n. This uses the true rows in the posterior: it holds for W_{unk} when rows are known ($r = 0$, the synthetic study) and for the joint coverage interval, which ranges over the true entries, when they are estimated. We do not claim convergence of $W_{\mathrm{unk}}(x)$ to a limit. (iii) is the stated policy.

Noisy labels on real data. Each reference label is flipped with known probability η to a uniformly chosen other source. With $T_l = \frac{1}{m}\sum_i \mathbf{1}\{\tilde{y}_i = l\} e_{\mathrm{cell}(i)}$, $\mathbb{E}T = MA$ with $M = (1 - \eta)I + \frac{\eta}{K-1}(\mathbf{1}\mathbf{1}^\top - I)$, invertible for $\eta < (K-1)/K$, and $\hat{A} = \Pi_\Delta(M^{-1}T)$ row-wise, where Euclidean projection onto the simplex cannot increase the distance to a simplex row. The error of T_l splits into cell sampling given which images landed under label l (McDiarmid over the n_l landed images) and the deviation of the landing counts $c_{jl} \sim \mathrm{Bin}(m, M_{jl})$, bounded by Bernstein's inequality and vanishing as $\sqrt{\eta}$. M^{-1} amplifies a row error by at most $\|M^{-1}\|_\infty$.

B MACHINE-CHECKED STATEMENTS

The supplementary Lean 4 project (Mathlib, pinned) checks the following; `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for each, and the file contains no `sorry`.

Lean theorem	Statement	Used in
`ball_contains_truth`	concentration event $\Rightarrow$ w^* satisfies the ball constraint	Lemma 1
`diam_bound`, `simplex_sub_sum_zero`	$\sigma\|w - w'\| \le 2\rho$ on the tangent space	Prop. 1
`ratio_lipschitz`, `posterior_lipschitz`	posterior moves by $\le 2a_{\max}\|w - w'\|_1/v$	Prop. 1, 3(i)
`two_point`, `width_ge`	same log law $\Rightarrow$ width $\ge \|\theta - \theta'\|$	Prop. 2
`novel_not_identified`, `novel_segment_in_simplex`	q in the hull $\Rightarrow$ non-identified segment	Prop. 3(ii)

Not formalised: the concentration inequality (Mathlib provides no McDiarmid inequality), the ℓ_1–ℓ_2 conversion, and the label-channel rates.

C IMPLEMENTATION DETAILS

Cells and head. Features are L2-normalised CLIP ViT-B/32 or DINOv2 ViT-S/14 image embeddings of bicubic-resized 224×224 images, extracted once and frozen. PCA to 32 dimensions and a pooled-covariance Gaussian discriminant with equal class priors are fitted on a fixed 10,000-image development split; the outlier cell collects inputs whose smallest Mahalanobis distance exceeds the development 99th percentile.

Radii and budget. $\alpha = 0.05$ is split equally between the log, the references and the label channel; the reference share is split over the $K + 1$ rows and the label-channel share over the $K \times d$ entries, each with a Hoeffding interval. The ball is solved without relaxation (second-order cone, Clarabel); the coordinate box outer approximation gives the same synthetic decisions but is about 30% wider there and close to vacuous with ten sources.

Baselines on real data. The constraint region follows the construction of Li et al. [2026]: weights in the simplex satisfying, for every cell, $\sum_j \underline{A}_j(x) w_j \le \hat{p}_n(x) + t$ and $\sum_j \overline{A}_j(x) w_j \ge \hat{p}_n(x) - t$, with t a Hoeffding half-width for the log frequencies (union over cells, budget $\alpha/3$) and $[\underline{A}, \overline{A}]$ the same per-entry intervals as REACH's joint interval. Posterior extrema over it are linear-fractional programs solved as LPs. It is our reimplementation from the paper's description, not the authors' code. The two-stage bootstrap resamples each reference row (m multinomial draws) and the log, refits EMQ, and takes percentile intervals (200 replicates).

Tight joint interval. With $u = tw$ and slack $s \ge 0$, each endpoint solves $\min / \max\ \mathrm{num}^\top u$ subject to $s_c \ge \sum_j u_j \underline{A}_j(c) - t\hat{p}_n(c)$, $s_c \ge t\hat{p}_n(c) - \sum_j u_j \overline{A}_j(c)$, $\|s\|_2 \le t\varepsilon_n$, $\mathbf{1}^\top u = t$, $\mathrm{den}^\top u = 1$ (Clarabel). Validity: on the event of Lemma 1 and the entry events the true marginal $\sum_j w_j^* A_j$ lies in the box, so its distance from $\hat{p}_n$ is at most ε_n. The constraint region of Section 5.3 uses per-cell intervals on the log (share $\alpha/3$) and the entry intervals (share $\alpha/3$); the tight interval spends the same $2\alpha/3$, while the joint interval also spends a third share on the row radius r. The tight interval is not provably nested in the joint interval (the entry box can have a larger Euclidean diameter than r); that it is never wider in our runs is an empirical finding.

Acquisition test (version 2). Pools are 50/20/30% log/label/evaluation, fixed per split. Component entries are floored at 10^{-4} before computing posteriors so the log-likelihood is finite. Shifted labels update rows as $(M_0 \hat{A}_k + \mathrm{counts}_k)/(M_0 + n_k)$. Ensemble MI uses 20 members that resample the log and the references; its tercile-to-action map is chosen among all 27 maps over three actions by calibration-seed regret. Evaluation risk is importance-weighted to the episode's prevalences. Because the prevalences and the clean rows depend only on the seed, episodes from different splits are not independent; intervals use a two-way cluster bootstrap over seeds and shift families.

Compute. The synthetic study runs on a laptop CPU in minutes. Feature extraction ran on one A10G GPU; the real audits and the blind-spot test ran on 32-core CPU containers, each in under an hour.

D ADDITIONAL RESULTS

Second backbone. With DINOv2 ViT-S/14 [Oquab et al., 2024] the head is 95.2% accurate and the resolution gate reads 1.22 (super4) and 1.75 (ten sources). 13 of 13 registered cells agree (12 of 13 with ten sources, the same saturated noise cell). The joint interval and the constraint region cover in every seed in every label-shift and unseen-class arm; at $n = 16{,}000$ their widths are 0.69 and 0.63 (super4; 0.86 and 0.72 with ten sources), and

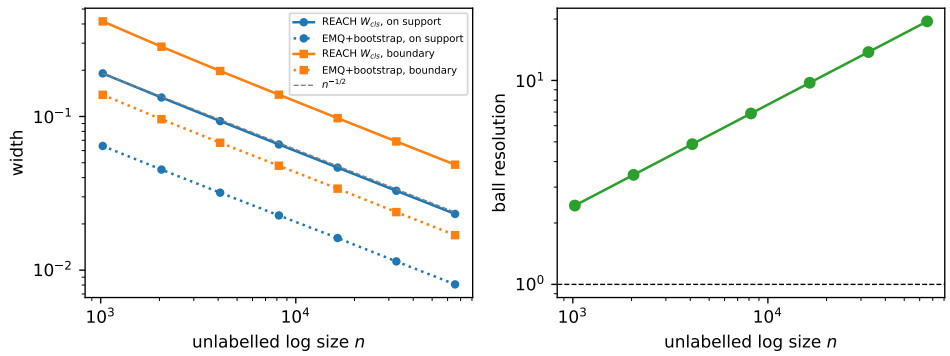


Figure 2: Synthetic, world *separated*. Left: W_{cls} (solid) and EMQ+bootstrap width (dotted) against n; dashed slope $-1/2$. Right: ball resolution, which must exceed 1 at the smallest n.

the two-stage bootstrap covers 0.74 (0.68). Plug-in W_{unk} covers 0.89 and 0.88 when unseen classes are present; the joint coverage interval covers in every seed. The corruption arm was not repeated for DINOv2.

Label noise (CLIP, super4). Over $\eta \in \{0.025, \ldots, 0.3\}$ the joint interval and the constraint region cover in every seed and both widen to 1.00 (joint $0.92 \to 1.00$); the two-stage bootstrap covers between 0.42 and 0.73.

Corruption–severity cells with joint coverage below 0.95 (CLIP; coverage among answered runs, and the rate at which the incompatibility test fires). *four superclass sources*: elastic transform 5 (0.74, flag 0.35); frost 5 (0.90, flag 0.40); glass blur 1 (0.70, flag 0.90); glass blur 2 (0.70, flag 0.90); glass blur 3 (0.70, flag 0.85); jpeg compression 4 (0.95, flag 0.05); jpeg compression 5 (0.87, flag 0.35); motion blur 4 (0.94, flag 0.00); motion blur 5 (0.85, flag 0.45); pixelate 5 (0.80, flag 0.75).

ten class sources: contrast 5 (0.80, flag 0.95); elastic transform 5 (0.92, flag 0.00); fog 5 (0.90, flag 0.00); frost 5 (0.94, flag 0.00); gaussian noise 2 (0.90, flag 0.00); gaussian noise 3 (0.81, flag 0.00); gaussian noise 4 (0.81, flag 0.00); gaussian noise 5 (0.80, flag 0.00); glass blur 1 (0.90, flag 0.00); glass blur 2 (0.90, flag 0.00); glass blur 3 (0.90, flag 0.00); glass blur 4 (0.86, flag 0.00); glass blur 5 (0.85, flag 0.00); impulse noise 4 (0.84, flag 0.00); impulse noise 5 (0.80, flag 0.00); jpeg compression 5 (0.94, flag 0.00); motion blur 5 (0.92, flag 0.00); pixelate 5 (0.93, flag 0.00); shot noise 3 (0.85, flag 0.00); shot noise 4 (0.83, flag 0.00); shot noise 5 (0.81, flag 0.00).

Where Proposition 3 stops: CIFAR-100. Known sources are four trees (maple, oak, palm, pine) and five vehicles, grouped into two sources; the novel source ($\delta = 0.1$) is willow, a sibling of the known trees, or whale. We registered that willow should show a smaller separation s, a flatter W_{unk} slope and a wider final W_{unk} than whale. With 400 reference images per class the novel row's radius alone is about 0.21 and the resolution gate passes only for $n \geq 8000$, so this arm is descriptive. With CLIP all three predictions hold ($s = 0.30$ vs 0.46; slope -0.02, inside the placebo band, vs -0.09; final width 0.57 vs 0.51). With DINOv2 only the slope prediction holds (-0.03 vs -0.06): in that encoder's cells willow is further from the known hull than whale ($s = 0.35$ vs 0.32). The joint novelty interval covers in all four runs. Whether a sibling class is inside the support is a property of the representation, not of the label hierarchy.

E PRE-REGISTRATION AND DEVIATIONS

Expected outcomes were committed in code before each test ran, and run manifests record each file's SHA-256. These are self-registrations in a local repository without third-party timestamps, so a reader can check consistency between files, hashes and results but not the order of events; we state the order below.

- Synthetic study: `train/expectations.py` (7a97650e…), before the first run.
- Real audit: `real/expectations.py`, in four states: 068095ec… (before the Dirichlet, label-noise and $\delta = 0$ runs, written after earlier fixed-skew runs had been inspected); 9a09b019… (adds DINOv2 and CIFAR-100,

before those features existed); 761495e4… (adds the CIFAR-100 grouping change and the joint coverage interval, after the first CIFAR-100 and DINOv2 runs); add1ccd3… (adds the two baselines of Section 5.3, after a one-seed smoke test, before the reported reruns).

- Acquisition test, version 1: `real/blindspot_prereg.py` (e699e72d…). Its header says 2026-10-03; it was written on 2026-10-02 at 19:32, one minute before the test code and after the corruption audit had shown the incompatibility flag rising with severity, so the off-frame rule was not blind to that link.
- Acquisition test, version 2: `real/blindspot_prereg_v2.py` (2fdecf48…), written 2026-10-02 23:09 after version 1's results and its review, before any version-2 code.
- Exchange-rate test: `real/exchange_prereg.py` (a70683e3…), 2026-10-03 06:52.

Deviations, all disclosed:

1. The coordinate-box ball was replaced by the exact ball after exploratory real runs were close to vacuous with it.
2. The joint interval was introduced after exploratory real runs showed plug-in W_{cls} under-covering, and the joint coverage interval after DINOv2 runs showed the same for plug-in W_{unk}. Both were then evaluated on the same seeds and the same test pool used in their design; their real-data coverage is therefore not a held-out confirmation.
3. A region-labelling error under grouped sources mislabelled the outlier cell; all grouped runs were repeated.
4. Coverage is judged at tolerance 10^{-6}, the solver's accuracy, after 10^{-9}-level solver output was seen to fail a 10^{-9} test where the exact extremum is 0. The synthetic study keeps the original 10^{-9} tolerance with the coordinate-box solver.
5. The first label-noise radius had an avoidable $\sqrt{K}$ factor and was replaced before the reported runs.
6. The CIFAR-100 grouping was changed after singleton sources failed the resolution gate at every n; the CIFAR-100 slopes are fitted over grid points where the gate still fails ($n \leq 4000$).
7. Acquisition test version 1 had two rule defects (a non-finite predicted log gain, when the hypothetical ball at the same histogram and smaller radius was empty, defaulted to labels; the label gain priced 25 clean labels per source while the action bought 100 shifted ones), tuned the baseline for matching accuracy rather than regret, and resampled (split, seed) episodes where its registration named seeds. Version 2 fixes all four; version 1 is superseded and its results are not used.
8. Corruption runs use 20 seeds per corruption and severity, not 50. Runs whose ball is empty abstain; coverage is reported among answered runs and unconditionally.
9. The version-2 acquisition rule, as registered, cannot recommend labels in a compatible frame (Section 5.4); we noticed this only after its run. We report the registered rule unchanged.
10. The third state of `real/expectations.py` carries a hand-written time ("∼23:25") that disagrees with its file time (23:12); the order relative to the runs is unaffected.
11. Exchange-rate test: `real/exchange_prereg.py` (a70683e3…) was written 2026-10-03 06:52, before its code and before any ResNet-50 feature existed. The first ResNet-50 run crashed in the ensemble baseline (every member's resampled log was incompatible with the estimated rows); the fix, which marks such episodes' MI as undefined and places them in the top tercile, was made before any ResNet-50 result existed. The synthetic and CLIP runs did not reach that case, and rerunning both with the final code reproduces their results (synthetic bit-exactly, CLIP to 10^{-15}). The crashed run's log is not released.